

Causal Importance Is Not a Property of the Latent

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Abstract

1 Introduction

A published causal-effect number for a sparse autoencoder (SAE) latent – Cho et al. [2607.20596], Marks et al. [2403.19647], Bricken et al. [2023], Templeton et al. [2024], Gao et al. [2406.04093] – is reported as if it were a property of the latent: ablate it, read a KL divergence or logit difference, publish the number. Table 3 audits these five papers, the ones we could confirm zero-ablate a single latent and read a magnitude rather than a match-rate accuracy or a multi-latent edit. None report dropping special-token positions before pooling. None report normalising the readout by the magnitude of the ablation. None report the near-duplicate structure of the dictionary they ablate. None report the position within the sequence at which the latent was measured, or whether that position generalises to others where the same latent fires.

This is not a checklist paper. The last of those four gaps turns out to be the largest one, and answering it changes the question. §4 shows that two of these choices are not merely omissions but can flip the sign of a real comparison on the same data. §5 then asks the question the audit table implies but no cited paper tests: if the same latent is measured under a different, equally defensible fitting choice for the dictionary it came from, does its causal effect rank survive? Six SAE arms sharing a single initialisation seed, varying only decoder freedom, learning rate, sparsity, and data order – exactly the choices a practitioner makes and a reviewer would accept – give latent identity 16.9% of the variance in log-causal-effect. Arm, latent $\times$ arm interaction, and position within latent give it the other 83.1%. Causal importance, in the sense the field currently measures it, is predominantly a property of the measurement, not of the latent.

2 Related work

Sanity Checks for SAEs (Korznikov, Galichin et al., 2602.14111) show frozen-random baselines match fully-trained SAEs on reconstruction, interpretability, sparse probing, and causal editing (0.73 vs 0.72). Their causal-editing metric is RAVEL’s interchange-intervention match-rate accuracy, not zero-ablation with a magnitude readout, so it is outside this paper’s scope (§11); we build our soft-frozen baseline (§3) as the construction they describe, but make no claim about their number. Their synthetic ground-truth result and lazy-training account are theirs and we do not reclaim them.

Are SAEBench Metrics Reliable? (Chanin et al., 2605.18229) find two SAEBench metrics, TPP and SCR, fail three independent reliability lenses. Both metrics zero-ablate a small *set* of latents per concept, not one, so they sit outside the audit table for a different reason than the RAVEL-based papers above – and TPP/SCR are also the only ablation-based metrics in SAEBench, so their failure is confounded with being the only set-ablation metrics; we do not attribute it to

the artifacts catalogued here. Their mechanism (reseed noise, low discriminability across training trajectories) is complementary to ours (directional confounds in a fixed measurement): together the two results say the ablation-based literature currently cannot distinguish a trained dictionary from a matched baseline with any of the tools on offer.

Are Single-Token SAE Features Causally Necessary? (Cho et al., 2607.20596) is the closest published methodology to ours and was investigated as a reanalysis target. Their headline depth correlation ($\rho = 0.81$, Table 21) survives a sensitivity bound on within-layer latent variance with room to spare; we did not find grounds to contest it. Independently, our own implementation of their single-token feature detector, run at their exact specification (layer 12, Gemma-2-2B), yields 0.41–0.44 $\times$ their reported prevalence at four times their apparent corpus size – a stable, unexplained reproducibility discrepancy we report as a finding about the detector, not a refutation of their causal claim.

3 Setup

We measure the causal effect of a single SAE latent by ablating its contribution to the residual stream and reading the change in the model’s next-token distribution. For latent f with decoder direction d_f and activation a_f at position t , we replace $h_t \leftarrow h_t - a_f d_f$ at the SAE’s hook point and record

$$\text{KL}_t = D_{\text{KL}}(p(\cdot \mid h) \parallel p(\cdot \mid h')) , \tag{1}$$

evaluated at position t and, for the propagation analysis, at $t+1$ and $t+3$. Experiments use `gemma-2-2b` and residual-stream Gemma Scope dictionaries, a matched trained/soft-frozen pair, and six SAE arms sharing a single seed (§5); a Gemma-3-1B replication is in progress at the time of writing.

Estimator. Group comparisons use the Hodges–Lehmann estimator, the median of pairwise ratios, reported with a bootstrap interval. HL is the point estimate belonging to the Mann–Whitney test; pairing a Mann–Whitney p -value with a ratio-of-medians interval, as we did initially, produces an apparent contradiction that is an artifact of mixing frameworks.

Convention. Ratios of the form $\text{KL}_{t+k}/\text{KL}_t$ are computed *feature-level*: a per-latent median first, then the ratio of medians across latents. The observation-level alternative differs by a non-constant factor 0.97–1.35 and is not used.

4 One comparison, two conclusions

Take the single comparison a practitioner would run first: does this dictionary’s rarest activation-frequency decile carry more or less causal mass than a frequency-matched decile of an untrained, tied-random dictionary? Under raw KL, the rare decile is *worse* than random (HL = 0.683, 95% CI [0.446, 0.962], $n = 40$ vs 35, Mann–Whitney $p = 0.039$). Under KL per unit perturbation norm, the identical latents and the identical control are *better* than random (HL = 1.687, 95% CI [1.205, 2.459], $p = 0.011$). Both intervals exclude one. Nothing about the sample, the model, or the dictionary changed between the two readings; only whether the intervention’s own size was accounted for did. Perturbation magnitude varies from 3.60 to 14.76 across activation-frequency deciles (§3 arms), so raw KL is comparing interventions of systematically unequal size and mistaking the result for a property of rarity.

component	2 arms (extrapolated)	6 arms (direct)
latent	–	16.9%
arm	0.4%	3.2%
latent $\times$ arm	9.7%	12.5%
residual (position)	–	67.4%
latent $\times$ arm : arm	23.6 $\times$	3.8 $\times$
$E\rho^2$ (one-arm design)	0.455	0.415
arms needed for $E\rho^2 = 0.8$	5 (extrapolated)	6 (direct)

Table 1: Crossed variance decomposition of log causal effect, latent $\times$ arm, method-of-moments ANOVA on a balanced subsample (200 latents $\times$ 6 arms $\times$ 6 positions per cell). The 2-arm column is the earlier trained-vs-soft-frozen design, with one degree of freedom on the arm factor; it is superseded by the 6-arm column, not reported as a separate result.

Corollary. Activation frequency predicts causal mass in neither the trained nor the control arm once magnitude is accounted for. The apparent frequency dependence in this setting is magnitude, not frequency – and it would have shipped as a rarity finding under the more common, uncontrolled convention.

5 Causal importance is not a property of the latent

The comparison in §4 shows that *how* a single fixed dataset is read can flip a conclusion. This section asks a different question: does the same latent, measured on dictionaries that differ only in a fitting choice, keep its rank at all?

Design. Six TopK SAEs share initialisation seed 0, so latent i denotes the same initial direction in every arm; each is then trained with one different, individually defensible choice: decoder free, soft-frozen at $\tau = 0.80$, soft-frozen at $\tau = 0.90$, learning rate $10\times$ lower, sparsity $k = 41$ instead of 82, or a reshuffled corpus order. Every arm sees the same 12M tokens. Live latents are intersected across all six arms (240 shared), one uniform sample is drawn from the intersection, and every arm is measured on exactly that sample at up to six positions per latent – giving a crossed latent $\times$ arm design with real replication on both factors, the direct six-level analogue of a model $\times$ wrapper design.

The headline survives going from one degree of freedom to five. The 2-arm design’s extrapolated $E\rho^2 = 0.455$ and its “5 arms needed” corroborate closely with the 6-arm design’s direct estimate ($E\rho^2 = 0.415$, 6 arms needed) – Table 1. This is not an artifact of a single contrast: latent identity accounts for only 16.9% of variance, and arm, latent $\times$ arm, and position jointly account for the other 83.1%.

What did not survive. The 2-arm latent $\times$ arm-to-arm ratio (23.6 $\times$) falls to 3.8 $\times$ with real replication on the arm factor. A single contrast between two arms is a noisy estimate of the arm component with one degree of freedom; six defensible fitting choices give arm a larger, more honest share (3.2% vs 0.4%). We report this reversal rather than quietly keeping the larger, earlier number: the ratio was never the load-bearing claim, and $E\rho^2$ is the number that held up.

Rank transfers partially, never fully. Pairwise Spearman correlation of each latent’s causal effect across all 15 arm pairs has median 0.376, range [0.225, 0.574] – every pair positive, none above 0.6. A latent that is causally important under one fitting choice is somewhat more likely to be important under another, but a one-dictionary measurement is closer to a single reviewer’s grade than to a latent’s property.

Robustness to the narrowest arm. The $k=41$ arm alone restricts the shared live-latent intersection to 12,080/16,384. Dropping it and recomputing on the remaining five arms changes every figure by at most 1.2 points (latent 16.9% $\rightarrow$ 16.8%, arm 3.2% $\rightarrow$ 2.0%, latent $\times$ arm 12.5% $\rightarrow$ 12.6%, residual 67.4% $\rightarrow$ 68.6%, $E\rho^2$ 0.415 $\rightarrow$ 0.412). The decomposition is not driven by the narrowest arm’s dictionary.

6 Position accounts for most of what is left

The residual term in Table 1 is position within latent: 67.4% of variance, larger than latent, arm, and their interaction combined. Of the five audited papers (Table 3), the ones whose methodology we could confirm in detail measure each latent at effectively one position per instance – a single curated token (Templeton et al.’s case-study ablations), the top-activating token (the convention we ourselves used before this decomposition), or an aggregate that is not itself decomposed by position (Gao et al.’s ablation-sparsity metric). None report what fraction of a latent’s own measured effect is explained by which position was chosen over another. At 67.4%, that choice explains more of the answer than which SAE the latent came from.

7 What a defensible measurement needs

Each item below is an artifact established in §8, stated as the correction it implies. A practitioner reporting ablation-based causal effects should:

1. **Drop special-token positions** before any pooled statistic. Including position 0 moves the explained variance of a released Gemma Scope dictionary from 0.863 to -3.5 .
2. **Report effect per unit perturbation norm**, not raw. §4 shows this alone reverses the sign of a real comparison.
3. **Report the per-latent nearest-neighbour cosine distribution**, and adjust or match on it *in dictionaries that exhibit near-duplicate structure*. Whether yours does is one line of code.
4. **Report unembedding alignment** alongside the effect. This is a sensitivity rather than a confound (§8), so both the unadjusted and alignment-adjusted estimates belong in the table.
5. **Report the jackknife curve**, not the mean or the variance ratio. Removing one latent of 240 moves our variance ratio from $42.1\times$ to $7.2\times$.
6. **State the sampling scheme**. The size of the alignment sensitivity depends on it, and no sample of a few hundred latents reaches the extreme tail of the alignment distribution.
7. **Report position, and how many positions per latent were measured**. §6: position accounts for more variance than which dictionary the latent came from.

8 Artifacts

Each subsection gives the mechanism, the evidence, the control, and the effect on the reported number.

8.1 Special-token reconstruction

Mechanism. The BOS residual carries a massive activation the dictionary does not reconstruct: cosine 0.44 between input and reconstruction at position 0 against ≈ 0.93 elsewhere. Its norm dominates the pooled variance. **Evidence.** Including position 0 gives explained variance -3.5 ; excluding it gives 0.863 at mean L_0 85.4, matching the released specification. **Control.** Drop special-token positions; assert the expected explained variance and abort otherwise. Without that assertion every downstream number is computed on a broken hook point.

8.2 Perturbation magnitude

Mechanism. Ablation perturbs the residual stream by $afdf$, and activation magnitude covaries with activation frequency, so raw effect sizes compare interventions of unequal size. **Evidence.** §4. Median perturbation norm runs 3.60 in the rarest frequency decile to 14.76 in the most frequent. **Control.** Report KL per unit perturbation norm throughout.

8.3 Near-duplicate structure

Mechanism. Feature splitting produces decoder rows at high mutual cosine; ablating one member of such a pair leaves the concept partly represented. **Evidence.** Per-latent maximum cosine predicts causal mass in the released Gemma Scope dictionary ($\rho = +0.220$, $p = 0.0006$) and in neither of our trained arms ($+0.078$, $+0.060$), which have visibly less splitting (medians 0.207 and 0.122 against 0.272). **Control.** One-sided adjustment where the relationship is asymmetric. Symmetric residualisation is wrong here: fitting the control arm produces a slope of $+2.506$ against a predictor with $p = 0.82$, i.e. residualising against noise. Matching is infeasible against a random dictionary, which structurally cannot contain near-duplicates.

This artifact is conditional. Near-duplicate structure confounds causal-effect measurement *in dictionaries that have it*. That is more useful than a universal claim, because it comes with its own diagnostic.

8.4 Direct-path alignment: a sensitivity, not a confound

Mechanism. A trained decoder direction is aligned with directions the model writes to the residual stream, and those feed the unembedding. Next-token KL therefore rewards latents that short-circuit the network. We measure alignment as $\sqrt{d^\top C d}$ with C the embedding covariance — the spread of the direct logit contribution, since a uniform logit shift leaves the softmax unchanged.

Evidence. Alignment predicts causal mass at $\rho = +0.375$ in Gemma Scope and $+0.264$ in a dictionary we trained ourselves with a different architecture and roughly $40\times$ less data, with the control arm null in both cases (-0.063 , $+0.004$). The single most causally powerful latent we found writes the letter k to the logits and carries $12\times$ the causal mass of the runner-up.

Why this is a sensitivity and not a confound. The trivially aligned latents are approximately the top 20 of 16,384, the 99.88th percentile. No sample a paper would realistically draw reaches them: the top twenty by alignment within a 240-latent sample sit near the 92nd dictionary percentile

and are 90% word-like, under both stratified and uniform sampling. So at realistic sample sizes high-alignment latents are semantic, and adjusting alignment away removes signal rather than triviality. The correct statement is that ablation-based measurements are sensitive to unembedding alignment at $\rho \approx 0.3$, and that within realistic samples this alignment reflects semantic content.

An unexplained depth dependence. Because C is layer-independent by construction, the measure estimates the logit effect a direction *would* have if it reached the output unchanged, so it should predict best where fewest blocks intervene — deepest. It does the opposite. Alignment predicts causal mass with $\rho = 0.358$ at layer 5 and 0.339 at layer 12, against 0.031 at layer 19 and 0.145 at layer 24 ($n = 300$ per depth).

At $n = 300$ a Spearman correlation of 0.3 carries a 95% interval of about ± 0.10 , so the two shallow estimates are indistinguishable from each other and so, marginally, are the two deep ones. We therefore test the trend rather than describe the sequence: pooling by Fisher z gives $\rho = 0.349$ for the shallow pair and $\rho = 0.088$ for the deep pair, a contrast of $z = 4.74$ ($p = 2.1e - 06$), and a weighted regression of z on layer index has slope -0.0163 per layer where the measure’s construction predicts a positive slope. The pattern is a decrease with depth, not a mid-network dip with recovery.

The compressed-variance explanation is falsified: alignment spread is equal across depths (CV 0.1175 against 0.1091) and causal-mass spread is *greater* at layer 19. We report the dependence as unexplained; it is a violation of the measure’s own construction rather than an observation about the model.

8.5 Position in sequence

Mechanism. Effects are measured at each latent’s top-activating positions, and later positions carry more context and a sharper next-token distribution. **Evidence.** Pooled $\rho = -0.087$ ($p < 10^{-4}$) across 2,700 latents, reaching -0.207 in one dictionary and null in five. The sign is *negative*: earlier positions carry slightly more causal mass, opposite to the mechanism we proposed. On the endpoint sample the effect is not significant ($\rho = -0.084$, $p = 0.196$). This correlational null does not contradict §6’s variance share: a latent’s effect can vary substantially across its own top positions (67.4% of variance) without that variation trending consistently early-to-late across latents ($\rho \approx 0$). **Control.** Record and report position, and how many positions per latent were sampled; a single position per latent cannot distinguish a stable causal effect from a coincidence of which token was chosen.

9 Results

9.1 Endpoints

Table 2 gives the trained-versus-control comparison for three controls. The second and third rows share a dictionary, a feature list and an evaluation, so their ratio is a within-experiment decomposition.

A trained dictionary beats an untrained one by $2.104\times$ and a soft-frozen baseline by $1.192\times$ ($p = 0.0213$). Most of the measured advantage is therefore attributable to the encoder; decoder freedom contributes the smaller factor. This holds despite the two arms differing sharply in decoder geometry — median cosine to initialisation 0.231 for the free decoder against 0.800 for the constrained one — and despite a reconstruction gap that grows monotonically over training ($+0.0064$ explained variance per million tokens, $p < 10^{-4}$).

comparison	HL	95% CI	n
Gemma Scope vs untrained tied	2.269	[1.934, 2.638]	240
ours vs untrained tied	2.104	[1.826, 2.423]	240
ours vs soft-frozen	1.192	[1.026, 1.379]	240

Table 2: Causal effect per unit perturbation norm, Hodges–Lehmann ratios on uniform samples of live latents. The soft-frozen control has a trained encoder and a decoder constrained to cosine 0.8 of its random initialisation.

9.2 The tail is not a population phenomenon

Across 9 Gemma Scope dictionaries spanning two depths, three widths and sparsity from L_0 22 to 445, the ratio of maximum to median causal mass within a fixed sample of 300 latents runs from $33\times$ to $464\times$, and the top 5% of latents carry between 36% and 73% of total causal mass. Every dictionary shows it.

But within any one dictionary the tail is a handful of latents. Removing the single largest takes the variance ratio from $42.1\times$ to $7.2\times$; the median ratio is unmoved. A bootstrap resamples those latents in and out while still treating them as population draws, so the jackknife curve is the honest instrument. At $n \approx 200$ the control arm’s 99th percentile has a bootstrap interval spanning $11.3\times$, and the derived “fraction above the control’s p_{99} ” runs [4.2%, 39.6%]: published aggregate comparisons on a few hundred latents cannot distinguish anything.

9.3 Effect decays with remaining depth, but not proportionally

Measuring (1) at t and $t+3$ across four depths at matched width and approximately matched sparsity gives KL_{t+3}/KL_t of 0.384 (layer 5, 20 blocks remaining), 0.360 (layer 12, 13), 0.078 (layer 19, 6) and 0.021 (layer 24, 1), with $n = 300$ latents per depth. The effect itself is large and not in doubt: a span of $18\times$ from shallowest to deepest.

The *attribution* to propagation limits is much weaker, and we state it as such. We pre-registered three predictions of decreasing strength. A power law in remaining depth, fitted from layers 12 and 19 and tested out of sample with a factor-of-1.5 tolerance, predicts 0.760 at layer 5 and 0.003 at layer 24 against observed 0.384 and 0.021 — misses of $0.51\times$ and $8.4\times$, in opposite directions, so the curve is flatter than proportional at both ends. Two threshold predictions, >0.5 at layer 5 and <0.02 at layer 24, also both failed. Only bare monotone ordering survived, and on four points that is worth $p \approx 0.042$ under a random-permutation null.

So: KL_{t+3}/KL_t declines with depth by a factor of 18; the propagation account predicts this ordinarily at $p \approx 0.042$ and is falsified as a proportionality in both directions. The decline is solid, the mechanism is not established.

Consequence for §8.4. Because this gradient is a propagation effect, it is not evidence that deeper layers are more direct-path dominated, and it must not be read as support for the alignment sensitivity.

9.4 Reconstruction improves without a matching causal gain

Evaluating the trained and soft-frozen arms at four training budgets gives HL ratios of 1.086, 1.098, 1.392 and 1.257 at 3, 6, 9 and 12 million tokens (slope $+0.0270$ per million, $p = 0.28$), while the explained-variance gap over the same range grows from 0.000 to 0.077 at $p < 10^{-4}$. The

reconstruction gap grows significantly and the causal gap does not. With four budgets, 120 latents each, and non-monotone point estimates, this is consistent with a dissociation but does not establish one.

10 Errors we made

We log these because each would have shipped as a real result if not caught, and because the reversals are themselves informative about where this kind of measurement goes wrong.

A sign error made a control column constant. Zeroing a self-similarity diagonal after taking an absolute value, rather than before, left every entry in one column equal to 2.0 instead of excluded. The near-duplicate analysis in §8 was rebuilt after this was caught.

Residualising against noise fabricated a confound. Fitting a covariate-adjustment regression on the random control arm – whose predictor has no true relationship to the outcome ($p = 0.82$) – produced a large, significant-looking slope (+2.506) that appeared to explain away a real effect in the trained arm. Adjustment must be asymmetric: fit only where the covariate relationship is real.

Pairing a Mann–Whitney p -value with the wrong interval created an apparent contradiction. A ratio-of-medians bootstrap CI containing 1 alongside a significant Mann–Whitney test looks contradictory because the two statistics belong to different estimators. Hodges–Lehmann is the point estimate the Mann–Whitney test actually tests; §3’s convention was adopted after this was diagnosed, not before.

A falsified prediction revealed the real mechanism. We predicted uniform (rather than frequency-stratified) sampling would show top-alignment latents to be geometrically trivial. It did not: wordlike fraction was 0.90 under both schemes. The real reason high-alignment latents look semantic is sample size, not sampling scheme – the top alignment percentile is unreachable at realistic n regardless of how you sample. The falsification is what produced §8.4’s actual argument.

An overclaimed headline was walked back to what survived. An early draft of §9’s depth result stated “monotonicity confirmed” as the primary finding. Of three pre-registered predictions, only the weakest (bare ordering, $p \approx 0.042$ on four points) survived; the power law and both thresholds were falsified. The corrected framing — decline is solid, mechanism is not established — is what appears above.

11 Limitations and conclusion

One model, for now. All measurements here use gemma-2-2b; a Gemma-3-1B replication of the six-arm decomposition is in progress. Whether the depth dependence in §8.4 is about relative depth or about this model is separately untested; Llama Scope dictionaries would answer that.

Training budget. Our trained arms see 12M tokens against Gemma Scope’s billions, and the reconstruction gap is still widening at that point, so any claim about what decoder training does is scoped to this budget. **One corpus** for the trained arms (WikiText-103), and one evaluation corpus.

Ablation only. These artifacts apply to metrics that zero or subtract a latent’s contribution and read a magnitude. They do not transfer directly to interchange-based scores such as RAVEL, which are match-rate accuracies; we make no claim about that result (§2). Ablation, interchange and clamping all perturb along the same d_f , so “use interchange instead” is not a remedy — the axis that matters is magnitude versus rate. **Attribution not established.** The two SAE Bench metrics that fail reliability auditing are the only ablation-based metrics in that suite, but also the only set-ablation metrics; the properties are confounded and we do not claim their failure follows from these artifacts. **Unexplained.** The depth dependence of the alignment sensitivity, and the sign

paper	readout	drops spe- cial tok.	norm. by magnitude	reports dupli- cation	accounts for alignment
Bricken et al. 2023	per-feature ablation; output logit change	—	—	broad, not tied to effect size	—
Marks et al. 2024	single-feature zero-ablation; KL divergence	—	—	—	—
Templeton et al. 2024	per-feature zero-clamp vs. linear attribution	—	—	broad, not tied to effect size	—
Gao et al. 2024	per-latent ablation; sparsity of effect vector (not magnitude)	—	different axis	—	—
Cho et al. 2025	single-latent ablation; KL/logit necessity score	this paper's Artifact 1	—	—	unclear vs. $\sqrt{d^\top C d}$

Table 3: Audit of the five papers we could confirm use single-latent zero-ablation with a magnitude readout. “—” means not stated in the sections we reviewed, not confirmed absent. Excluded as a different readout family: Makelov et al. 2024 (multi-feature edits), SAEBench TPP/SCR (set ablation), Korznikov/Galichin et al. 2026 (RAVEL match-rate), Rajamanoharan et al. 2024 (whole-dictionary substitution) – see §2.

of the position-trend effect in §8 (distinct from the position-*variance* result in §6, which is about spread within a latent rather than a trend across latents).

Five audited papers report a single-latent causal effect with a magnitude readout; none report the position, duplication, or magnitude controls this paper establishes as necessary, and a sixth family of closely related papers (RAVEL-based, set-ablation) uses a different readout entirely for reasons we make explicit rather than dispute. Two of the controllable choices flip the sign of a real comparison (§4). A crossed six-arm design – the first, to our knowledge, to give the arm factor real replication in this setting – shows that a latent’s own identity accounts for 16.9% of the variance in its measured causal effect, and that which position of that latent’s own activations was measured accounts for more than that. Causal importance, as the field currently measures it, is a property of the measurement first and the latent second.